```
syms dw1 dw2 dw3 Omega I1 I2 I3

% spin about b1
I_mat = diag([I1, I2, I3]);
omega_vec = [dw1; dw2; Omega + dw3];
H_b = I_mat * omega_vec;

pretty(H_b);
```

$$\begin{pmatrix} I_1\,\mathrm{dw}_1 \\ I_2\,\mathrm{dw}_2 \\ I_3\,(\Omega+\mathrm{dw}_3) \end{pmatrix}$$

```
dw_i = I_mat \ cross(-omega_vec, H_b);
pretty(dw_i);
```

$$\begin{pmatrix} \dfrac{I_2\,\Omega\,\mathrm{dw}_2 - I_3\,\Omega\,\mathrm{dw}_2 + I_2\,\mathrm{dw}_2\,\mathrm{dw}_3 - I_3\,\mathrm{dw}_2\,\mathrm{dw}_3}{I_1} \\ -\dfrac{I_1\,\Omega\,\mathrm{dw}_1 - I_3\,\Omega\,\mathrm{dw}_1 + I_1\,\mathrm{dw}_1\,\mathrm{dw}_3 - I_3\,\mathrm{dw}_1\,\mathrm{dw}_3}{I_2} \\ \dfrac{I_1\,\mathrm{dw}_1\,\mathrm{dw}_2 - I_2\,\mathrm{dw}_1\,\mathrm{dw}_2}{I_3} \end{pmatrix}$$

```
dw_i = subs(dw_i, [dw1*dw2, dw1*dw3, dw2*dw3], [0, 0, 0]);
pretty(dw_i);
```

$$\begin{pmatrix} \dfrac{I_2\,\Omega\,\mathrm{dw}_2 - I_3\,\Omega\,\mathrm{dw}_2}{I_1} \\ -\dfrac{I_1\,\Omega\,\mathrm{dw}_1 - I_3\,\Omega\,\mathrm{dw}_1}{I_2} \\ 0 \end{pmatrix}$$

```
dw_i = subs(dw_i, [I1, I2, I3, Omega], [120, 100, 80, 1])
```

```
dw_i =
```

$$\begin{pmatrix} \dfrac{\mathrm{dw}_2}{6} \\ -\dfrac{2\,\mathrm{dw}_1}{5} \\ 0 \end{pmatrix}$$

```
pretty(dw_i)
```

$$\begin{pmatrix} \frac{dw_2}{6} \\ -\frac{2\,dw_1}{5} \\ 0 \end{pmatrix}$$

```
dw_i = simplify(dw_i);
zeroRows = all(isAlways(dw_i == 0), 2);
```

```
Warning: Unable to prove '-(2*dw1)/5 == 0'.
Warning: Unable to prove 'dw2/6 == 0'.
```

```
dw_i = dw_i(~zeroRows, :);
pretty(dw_i)
```

$$\begin{pmatrix} \frac{dw_2}{6} \\ -\frac{2\,dw_1}{5} \end{pmatrix}$$

```
x2 = [dw1; dw2];
A = jacobian(dw_i, x2)
```

A =

$$\begin{pmatrix} 0 & \frac{1}{6} \\ -\frac{2}{5} & 0 \end{pmatrix}$$

```
[V, D] = eigs(double(A))
```

```
V = 2×2 complex
     0.0000          -     0.5423i      0.0000          +     0.5423i
     0.8402          +     0.0000i      0.8402          +     0.0000i
D = 2×2 complex
     0.0000          +     0.2582i      0.0000          +     0.0000i
     0.0000          +     0.0000i      0.0000          -     0.2582i
```

```
V = V ./ V(1,1)
```

```
V = 2×2 complex
     1.0000          +     0.0000i     -1.0000          +     0.0000i
     0.0000          +     1.5492i      0.0000          +     1.5492i
```

```
D
```

```
D = 2×2 complex
     0.0000          +     0.2582i      0.0000          +     0.0000i
     0.0000          +     0.0000i      0.0000          -     0.2582i
```